Desktop XY Plotter Based Pick-and-Place and In-Circuit Test System

ELE 495 – 1st Progress Report

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Group 3

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Abstract

This project targets a low-cost desktop pick-and-place platform built on a cartesian XY plotter with an additional Z axis, a vacuum pickup head, and a top-mounted USB camera. The system will place through-hole and SMD components (initial focus: 2512 resistors and diodes) on a PCB fixture and will verify placement correctness via post-placement vision-based measurement.

The system architecture separates deterministic motion/IO (ESP32) from higher-level orchestration, vision processing, and web-based user interface (Raspberry Pi 4). The Raspberry Pi communicates with the ESP32 over USB serial. A web UI will allow the operator to define the placement sequence and monitor status, while manual intervention will be minimized.

Literature Survey

Open-source ecosystems demonstrate that a modular architecture (controller + firmware + machine) and camera feedback are common design patterns for desktop pick-and-place systems. For example, OpenPnP describes an open-source SMT pick-and-place system and documentation covering job execution, calibration, and camera-assisted workflows [1][2].

LitePlacer is frequently referenced as a low-cost prototype pick-and-place machine that can be operated with dedicated software or controlled by OpenPnP, indicating the viability of kit-based hardware combined with configurable software stacks [3].

In vision-guided assembly, camera calibration and coordinate-frame alignment are highlighted as critical to achieving reliable accuracy and repeatability. Applied research emphasizes that intrinsic/extrinsic calibration and robust reference-frame transformations are essential for accurate pick-and-place and inspection tasks [4].

Compared to typical hobbyist systems, this project integrates explicit post-placement accuracy measurement and reporting (camera-based verification), and enforces a split-controller approach (Pi for high-level logic and vision; ESP32 for deterministic motion).

Method

System Overview

Figure 1 illustrates the proposed system architecture. The web UI is the operator-facing layer for job definition (component sequence) and runtime monitoring. Raspberry Pi 4 performs job orchestration, vision processing (camera calibration, fiducial/localization, and post-placement measurement), and communication bridging. The ESP32 firmware provides deterministic generation of STEP/DIR signals for stepper motors (X/Y/Z), homing via endstops, and vacuum/valve IO control.

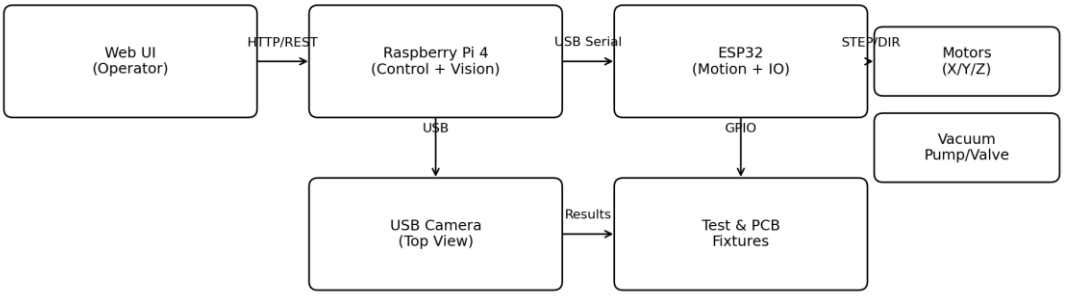

Figure 1. System block diagram

Functional Workflow

1) Homing: ESP32 performs a safe homing routine using Xmin/Ymin/Zmin endstops.
2) Job loading: The operator defines the placement order in the web UI; the Raspberry Pi converts the job into motion and IO commands.
3) Pick: The head moves to the component pickup location; the vacuum is enabled.
4) Place: The head moves to the target coordinate and places the part with controlled Z motion.
5) Verification: The top camera captures an image; the Raspberry Pi estimates placement offsets relative to reference features and reports errors to the UI.

6) Logging: All commands and verification results are logged (GitHub repository maintained by Arda).

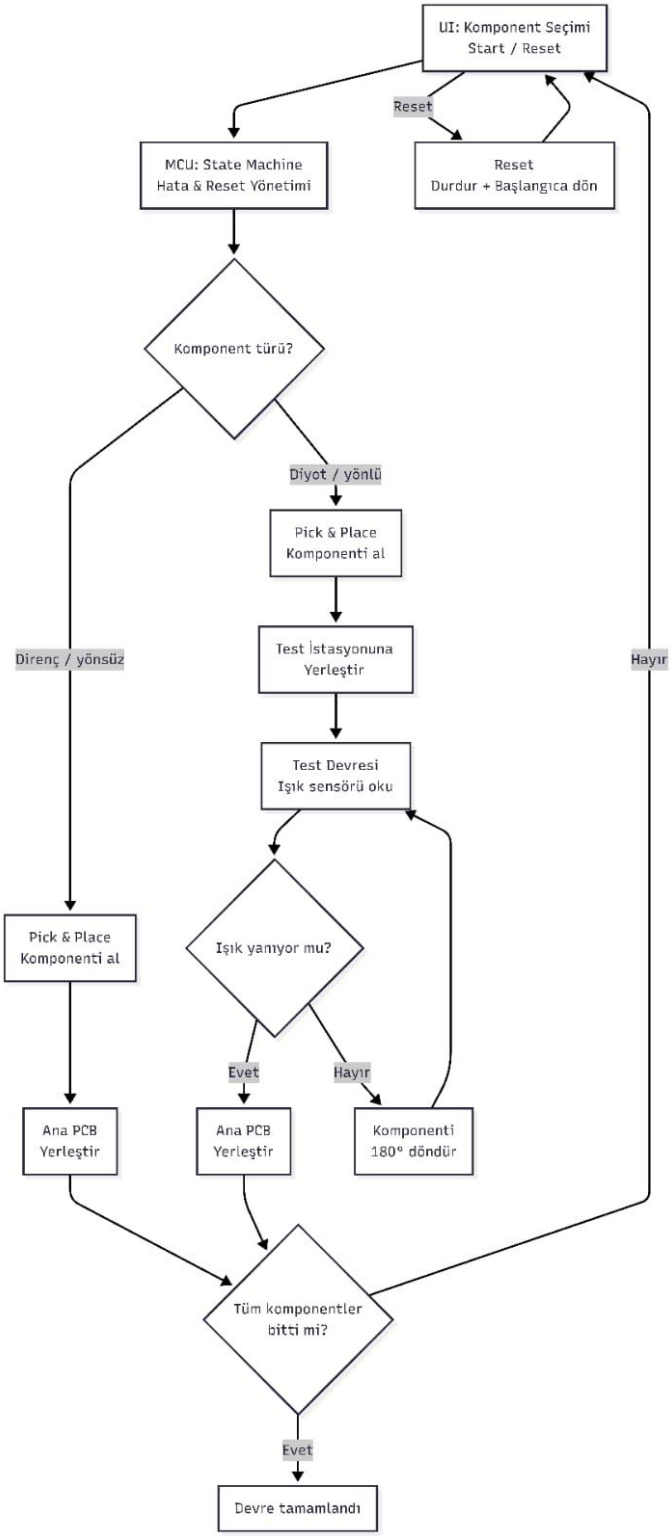

Figure 2 Function Block Diagram

Key Design Requirements

- Total project budget cap: 20,000 TL.
- Raspberry Pi 4 as the high-level controller; ESP32 as the real-time motion/IO controller.

- USB serial communication between Raspberry Pi and ESP32.
- Desktop XY plotter mechanics with an added Z axis.
- Endstop-based homing on all axes.
- Vacuum pump based pickup head; minimal manual support during placement.
- Top-mounted fixed USB camera for verification (no moving camera).
- Web-based UI to define placement sequence and display status/results.
- Post-placement accuracy measurement and reporting (e.g., mean/max error per job).
- Team-wide report ownership; testing and final demo readiness.

Planned Tests and Performance Metrics

Planned testing includes: (i) axis repeatability checks, (ii) homing repeatability, (iii) placement accuracy for 2512 resistors/diodes measured by the top camera, and (iv) end-to-end job execution reliability. The primary metric will be placement error (in mm) computed from camera measurements relative to expected footprints; results will be displayed on the UI and stored for traceability.

Budget

Table 1 lists major parts planned for procurement and integration. Unit costs and invoices will be finalized in the accompanying Excel sheet, and invoices/price proofs will be appended as required.

Item	Qty	Unit Price (TL)	Total (TL)	Status
Raspberry Pi 4 Model B	1	3314	3314	Purchased
ESP32 development board (USB serial)	1	248	248	Purchased
Stepper motors (X/Y/Z)	3	229	689	Planned
Stepper drivers	3	142	426	Planned
Endstop switches	3	20	60	Planned
Vacuum pump + valve + tubing	1	280	280	Planned
USB camera (top view)	1	956	956	Purchased
Power supply & wiring	1	100	100	Planned
Mechanical parts (rails/belts/idlers)	1 set	4800	4800	Planned
Custom PCB (control/IO)	2	502	1004	Planned
Extra outcomes			2000	Unplanned

Table 1. Budget and procurement list

PROJECT PLAN (WBS and GANNT Chart)

The project plan is presented with a Work Breakdown Structure (WBS) and an 8-week Gantt chart. Figures 3 and 4 are referenced throughout the plan.

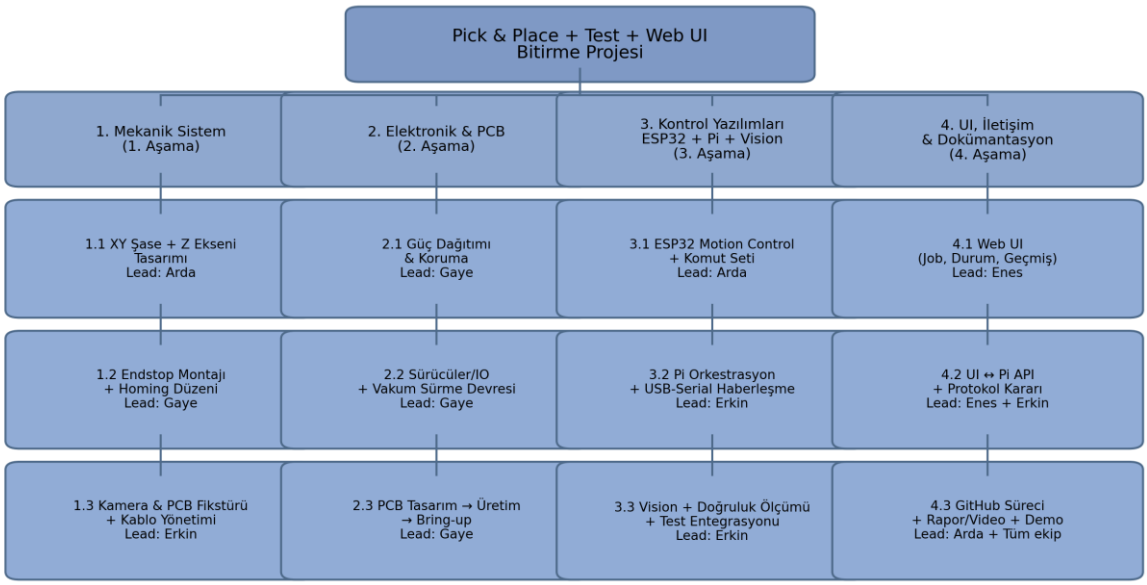

Figure 3. Work Breakdown Structure (WBS)

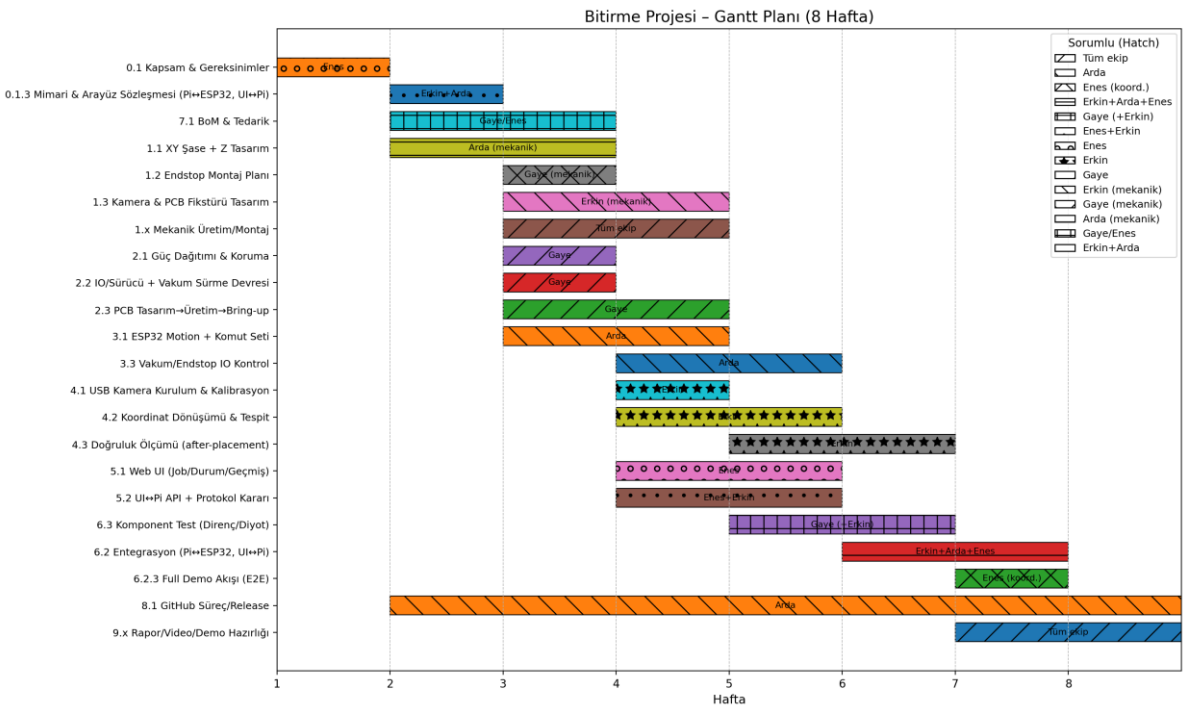

Figure 4. Gantt plan (8 weeks).

Work Package Ownership Summary

Electronics/PCB: Gaye Erdoğan. Raspberry Pi software & vision: Erkin Coşkun Ruhi. ESP32 firmware & motion control: Arda Aslangül. Web UI and operator communication: Enes Yeğin. Mechanical design and build tasks are shared across the full team.

What Will We Gain

At the end of this project, we will gain three concrete outcomes:

- **A working desktop pick-and-place + test prototype**: a functional XY+Z mechanism with vacuum pickup, endstop-based homing, and an operator-driven placement flow that completes a full job with minimal manual intervention.
- **An end-to-end software and control stack**: an architecture where the **ESP32** executes deterministic motion/IO (STEP/DIR, homing, vacuum control) and the **Raspberry Pi** runs orchestration, vision-based verification, and a **web UI** for job definition, status monitoring, and reporting.
- **Validated engineering competence and reusable assets**: we will leave with measured performance data like repeatability, placement error, test results, documented design decisions mechanical, electronics, firmware, UI/protocol, and a version-controlled repository code + documentation that can be extended for future work more component types, better calibration, faster cycle time, richer inspection/test coverage.

References

[1] OpenPnP. (n.d.). OpenPnP – Open Source SMT Pick and Place. Retrieved January 22, 2026, from https://openpnp.org/

[2] Dickinson, T. (Ed.). (2025, July 31). OpenPnP User Manual (OpenPnP Wiki). GitHub. Retrieved January 22, 2026, from https://github.com/openpnp/openpnp/wiki/User-Manual

[3] markmaker. (2024, February 1). LitePlacer (OpenPnP Wiki). GitHub. Retrieved January 22, 2026, from https://github.com/openpnp/openpnp/wiki/LitePlacer

[4] Parpală, R. C., Ivan, M. A., Parpală, L. F., Coteț, C. E., & Popa, C. L. (2024). Camera Calibration in High-Speed Robotic Assembly Operations. Applied Sciences, 14(19), 8687. https://doi.org/10.3390/app14198687